Curated to Win: Certifying Reward-Model and Judge Selection Against Evaluation-Set Curation

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Problem

Research pipelines need an evaluator: a reward model or an LLM judge. They pick one by its accuracy on a validation set, but the pipeline itself often decides which items go in that set.

How much of the validation set must be removed before a different evaluator wins?

Example: RewardBench

Model	All items	Without 12 items
INF-ORM-Llama3.1-70B	95.11	95.05
QRM-Gemma-2-27B	94.44	95.11
Skywork-Reward-Gemma-2-27B-v0.2	94.26	94.80

Removing 12 of 2,985 items puts the runner-up first (scores from RewardBench's own scoring code).

Setup

Model *m* has official score

$$S_m = \sum_i w_i^0 \, r_{mi}$$

A **curation** keeps a subset *K* and rescores every model on it. *K* **crowns** *m* if *m* is strictly first on *K*.

$$S_m^K = \frac{1}{f} \sum_{i \in K} w_i^0 \, r_{mi}, \qquad f = \sum_{i \in K} w_i^0$$

Certificate in weight

Let $g_c = S_a - S_c$ be a challenger's gap to the winner *a*. Any curation that lets *c* tie or beat *a* removes at least g_c:

$$S_c^K \geq S_a^K \implies 1 - f \geq g_c$$

$$f\,(S_a^K - S_c^K) = g_c - \sum_{i \notin K} d_i^c \;\geq\; g_c - (1 - f)$$

So the winner is safe against any removal below its gap to the runner-up, $g^\star = \min_c g_c$.

Certificate in items

c overtakes *a* exactly when the removed contributions sum past the gap:

$$d_i^c = w_i^0 \, (r_{ai} - r_{ci})$$

$$S_c^K > S_a^K \iff \sum_{i \notin K} d_i^c > g_c$$

Sort the contributions and count. This is exact: any $k^\star - 1$ items are safe, and some $k^\star$ items change the winner.

$$k_c = \min\Big\{k : \sum_{j \leq k} d_{(j)}^c > g_c\Big\}, \qquad k^\star = \min_c \, k_c$$

Data and method

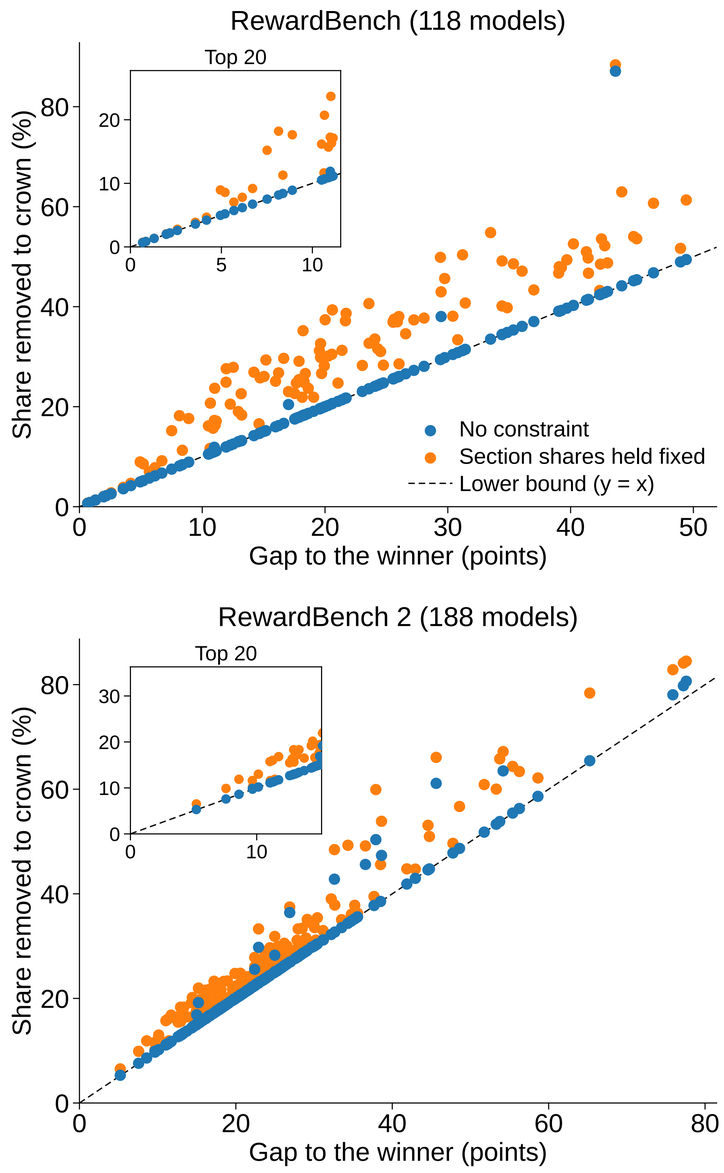
RewardBench has 118 models and 2,985 items. RewardBench 2 has 188 models and 1,753 items in five accuracy subsets (Ties excluded; adding it keeps the same winner). All results come from public per-item files.

For each model, a linear program over item weights searches for the cheapest curation that crowns it. A rounded item subset counts only if the model is then strictly first, so each cost is a verified upper bound.

Results

	Reward-Bench	Reward-Bench 2
Top-two gap $g^\star$ (points)	0.67	5.22
Cheapest curation found (% of weight)	0.68	5.30
Items that change the winner, $k^\star$	12	43
Same, under the benchmark's own code	11	40

How tight is the weight bound?



Every model can be crowned. Removed share over gap has a median of 1.00 on both benchmarks and is at most 1.02 in RewardBench's top 20 (largest overall: 2.00 and 1.35). Holding section shares fixed costs a median of 1.39 and 1.12 times the gap.

Possible winners

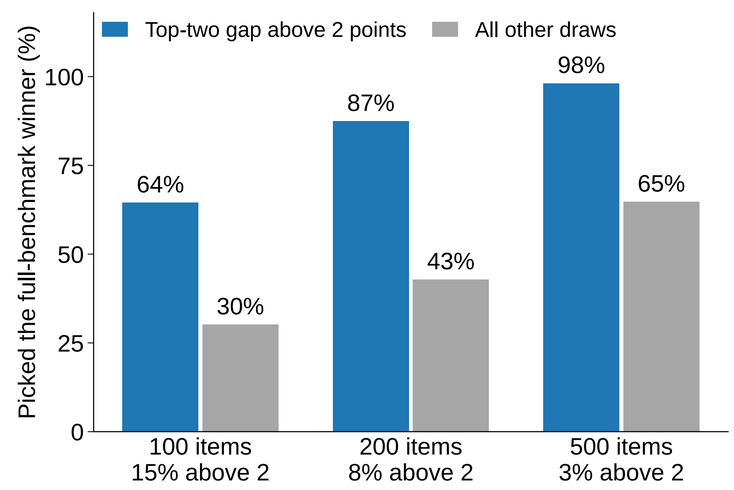
If up to a share δ of the benchmark may have been curated, how many models could be the winner?

Budget δ	0.5%	1%	2%	5%	10%
RewardBench	1	3	5–6	11	19
RewardBench 2	1	1	1	1	6

5–6 means 5 models have a verified curation and 6 are within δ of the winner.

Small validation sets

On a small set, the certificate is its top-two gap. Random RewardBench sets, 2,000 draws per size:



These draws measure sampling noise, not curation.

Protocol

1. Fix the items and record a hash before scoring. Score every candidate on all of them, and log every later filter.
2. Set a curation budget δ.
3. Report $g^\star$ and $k^\star$. Accept the winner only if $g^\star > \delta$.
4. Otherwise, report the set of possible winners.

RewardBench at 1%: $g^\star = 0.67$ points, so report 3 possible winners. None of this needs extra model calls.

Limitations

- Removal only, not reweighting or paraphrasing
- Weighted accuracy only, not mean win rate
- LP costs are upper bounds; the item count is exact
- I do not claim any published result was curated

Code and derived data

github.com/adrian-erlikhman/curated-to-win

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